

E-Mail Phishing Prevention and Detection

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ABSTRACT

Phishing is basically the type of cybercrime in which attackers imitates a real person through institution and mimics that they are sending message from an authorized organization and then take the details of the user (personal identity, credit card details and any type of bank information) and will breach the personal details of the user. There are many free tools to help in web based scams. Basically the free anti-phishing toolbars in the below given study were examined many example- in which Spoof Guard anti phishing toolbar is sufficient and good at identifying fraudulent sites and can also gave false positive results. Earth Link, Google, Net Craft, Cloud Mark and Internet Explorer seven detected many of the fraudulent or fake sites even more than 15% of fraudulent sites are false positive. Trust Watch, eBay and Netscape correctly found the fraudulent websites and by the combination of the toolbars the expected outcome came out.

KEYWORDS: E-Mail Prevention, Methods, Detection, Conclusion and Detection

1. INTRODUCTION

Phishing is a type of social engineering attack which is use to steal data or we can also say that phishing is basically the performance of breaching (leaking the personal information of the user) and attempt to breach or leak the personal details of the user in order to do fraud or to take the bank card details and of the person or user. Information can be – username, password and credit card details. Phishing mostly happened through popular websites and links, online payment processor IT administrator which is commonly used by public domains. This occurs when an attacker or phisher pretends to be trusted entity in order to dupe up a victim or user on to clicking a malicious link which will lead to installing of malicious malware, and which will stop dead the system which is a fragment of ransom ware charge and will reveal the sensitive information of the user. Phishing is one of the oldest and the type of cyber-attacks, since from 1990s. and it is one of the most widespread and damaging cyber-attack. There are two main consequences of phishing attack –financial loss, data loss and legal lawsuits. Phishing contains fake emails, websites and links.

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Types of phishing: - Phishing has been spread via mails and fake links to include VOIPS, SMS.

➤ **Clone phishing:** It is a type of attack where phisher or attacker tries to clone the website where user is visiting. The cloned website asked user to login credentials, copying the real websites. This will give authorization to the attacker to save the details of the user or victim. While clone phishing uses a legitimate or previously sent mails which contains the attachment and links. It is almost the real copy of the web site or the email. The email is spoofed by the attacker and originally looks that the real organizations sent the mail and the attacker will wait for the victim who will click it. When the user will click the link, the attacker will scam or clone the link and attacker will forward the same forged link or email to the contacts of the victim inbox. Clone phishing attack is considered as a most harmful and dangerous because it is hard for victim to recognize the spoofed email.

➤ **Spear phishing:** Spear phishing is basically the group specific targeted attack; phishers use to target selected groups instead of casting out of

thousands emails randomly. And in spear email the common things are there in targeted organization of group of people for example- www.ugdg.co.uk.in add another link will also similar like above one like- www.ggoogle.co.uk.in in both of the links the co.uk.in are common in a group of people. The spear phishing attack is the highly targeted form of phishing attack. It is very hard to recognize the spear attack. Criminals use to personalize the emails and impersonate the specific sender. The mail goal of the attacker or criminal is to trick target in to clicking a link or opening an attachment. Spear phishers use to target the specific individuals within the specific organization but with the specific mission. Spear phishing is basically the most prevalent delivery method for (APT) which is advanced persistent threat attack. Today the cyber criminals and Governments are launching the APT attack with sophisticated malware in order to detect the attack and to get inside the personal account of the user. Spear phishing is presented by the one or more of the following characteristics.

- **Use of zero- day vulnerability:** It is advanced spear phishing attack which leverage zero - day vulnerabilities in browser and also in plug –in or desktop applications in order to compromise the system.
- **Blended or multi vector threat:** In spears phishing the criminal uses blend of emails spoofing or dynamic URLs.
- **Multistage attack:** It is the initial exploit of the system of the first stage of APT attack which involves the other stages of the malware out bound communication.
- **Well – crafted emails forgeries:** In this attacker use to threat the targeted individual with specific organization.
- **Phone phishing /Vishing :** This is the type of phishing which refers through messages and phone calls in which the criminal or attacker use to pretends that they are bank workers or manager and asking the user to telephone or call up 1 a number concerning the funds of the user or details or in simple terms we can also define that the phone phishing refers to the phone calls from the people who pretend or claim to be from authorized and well reputable company. In order to get the details of the victim the criminals use to some social engineering tactics like psychological and social method of manipulating or tricking the user. Vishing can be performed in several ways like hybrid calls which is the hybridization of two

parties. Most of the phishing calls are made by using voice over internet protocol technology which is conducted with the caller ID“spoofing”.

There are major common 4 types of examples of phone phishing:-

1. **Telemarketing fraud:** This type of phone phishing attack is encompassing many types of phone spam calls for example- in the name of charity they will ask you to help out the people with fund and also asks for the bank account details or phone call may be also on that you have won the lottery in which for deposition of lottery amount criminal will ask you to give the information.
 2. **The internal revenue service:** It involves the malicious attack which pretends that works with at the IRS where the attacker will ask you to owe the taxes and to pay this immediately.
 3. **Medicare:** It is the type of scam which included the calls from the healthcare department and pretending to work for medicine.
 4. **The social security administration:** In this the criminal is pretending to be from the social security administration (SSA) and will gather each facts and details from the holder of the account and asks the holder about the security benefits.
- **DNS based phishing:** DNS basically stand for Domain Name Server. DNS based spoofing is the type of a charge which changed the details which are used to deflect or swing online trade to a fake websites and links. Once the user logs in their own verified account, the criminal will take the advantage and will steal the all personal and sensitive information of the user even, name passwords to the bank account details. DNSSEC is the protocol which is designed to secure the DNS by verification method. Where the method uses the unique which is called as cryptographic signatures which are stored alongside with the DNS records.

DNS can help to protect against the following terms:

1. **Absence of data hiding:** DNSSEC gives permission but it fails to cipher or encrypt the DNS retaliation due to which the attacker is adequate attention on congestion and uses that information for more experienced attacks.
2. **Zone enumeration:** DNEESC enables the signature validation. It can also be used to substantiate the negation of a DNS zone.

3. Complex deployment: DNEESC is also mis-configured and that could source server to drop the surety welfare and also refuse the authorization to a websites completely.

➤ **Man in the middle attack:** It refers to an attack where the criminal or attacker puts himself or herself in between the user and there cognized websites. The criminal also uses the malware to open the communication channels in between the user and the sites. Man in the middle attacks can be used to execute advanced persistent threat (APT). For example—in an HTTP transaction, a TCP connection exists between the user and the server, the attacker will split the TCP transfer control protocol connection in between two one is for the attacker and the other is between the attacker and the server.

➤ **Whaling:** These types of attack are more targeted and the main aim is same as just like phishing. Where criminal attempt to create or mimic the senior executives by some tricks which are fake links and websites or malicious URLs. Attackers in whaling target the high profile employees. The main goal of the attackers is to manipulate the victim in to authorized high value wire transfers to attackers. Due to highly targeted nature whaling attacks are very difficult to detect and to prevent standards.

Theoretical aspects of phishing techniques:

➤ **Email spoofing—** Is used to make the fraudulent email activity hiding email origins. Basically, email spoofing occurs when the attackers are able to alter the emails senders' information. Email spoofing is done by basically spammers. Phishing email contains hostile reasons similar like a virus escalates in order to obtain the bank account details. SMTP (is basically the simple mail transfer protocol) which fails to layout, somewhat kind of confirmation procedure for forwarding emails.

➤ **Spoofing utilize the exhibit title artifice:** It is the majority ordinary form of email spoofing and is very triumphant since majority of email clients show only the display name directly. With this entity the attacker will insert the identity of the trusted brand or targeted company into front name. This name or type of deception attack is easy economic.

➤ **Spoofing is also done by using legitimate domains:** With inclusion to influence of the front line name, sometimes phisher can work with the authenticate email address of the targeted specification in the form of caption e.g., “United

Customer Service”. This type of deception is basically called as the estate spoofing deception or fraud. Email authentication standards such as DMARC can be used by the estate holder in order to stop spoofing from the deception.

➤ **Spoofing using looks alike domain:** When the domain is protected by the email authentication in that case it is very difficult to spoof the domain name. Attackers are used to or attempt the cheat the holder by record and utilize the domain name alike or closed to impersonate domain. And these types of attacks, known as looks-alike domain attacks. For example, attacker uses the rendering similarities, such as “PayPal” which reveals the particular type face and the way of the manner in most common electronic mail holder or user. Different type of the lookalike deception is adding the disposition from different script in the different sets of code for example from header "Drop Box" notification@dropbox.com= ">, where the 'o' is the cylicric characters.

➤ **Email spoofing and business compromise:** Comparing last two decades email spoofing and phishing attack has been increased in business. According to recent study by the “Agari cyber intelligence” the strategy is additionally used to appeal gifts boards for welfare funding to recompense their personnel works.

➤ **Stop electronic - mail spoofing in in case:** This is very impossible facing to stop the email parody from the cyber criminals because attackers are always discover different methods to device their target and to attack on the system of the user. Secure email cloud ensured that targets safety of the user from brand impersonation, spoofing etc.

➤ **Web spoofing:** Web spoofing are the most common attacks which are increasing day by day from past recent years. These attacks are easy because of two reasons; they are easy to execute and they work. Web spoofing is very difficult to identify because the fraud takes place outside the organization security and parameters. A phisher can forget the website which looks identical to the original websites just to pretend to the user that this site is original. Web spoofing creates the original copy of the World Wide Web.

Modern web browsers have security indicator which includes the domain name highlighting and HTTPS indicators. Modern web browsers display a pad lock icon when the user will visit an HTTPS web which is hypertext transfer protocol which is known as transport layer security which provides encryption and the identification.

- **DNS cache phishing:** DNS domain name server spoofing is an attack which alters the DNS records. The reason for this attack is making request to DNS resolver and then forging the reply and it also cause the name server to return in correct result record e.g., IP address.
- **Malware:** It is a software which is use to disrupt computer operation and the sensitive information. It can be in the form of scripts, code, active content and the other software's. Malware includes viruses, worms, key loggers, spyware, Trojans and adware.

Anti-phishing techniques:

Anti-phishing tools are based on the variations of the procedures that can be used to recognize the tools are integrated in to the web browsers that is depicted web pages which are phished including white lists, blacklists and community ratings. Anti-phishing tools are integrated into the web browsers that are depicted.

Anti-phishing techniques are majorly classified into four categories which are as follows-

- **Content filtering:** In this method the email and the spoofed content are filtered when it get enters in the victim mail box using the few methods e.g., support vector machines.
- **Blacklisting:** It is basically the collection of blacklists known phishing websites which are published by trusted entity for example Google and Microsoft blacklist.
- **Symptom-based prevention measures:** It is the analysis of the content of each web page which is visited by user which produces phishing alerts that what type of indications which are to be detected.
- **Domain binding:** Client browser-based technique where the sensitive information is bind with the particular domain. Domain binding alerts or warns the user when the user visits the domain.

Caller ID Toolbar:

Caller ID toolbar use variation in analysis methods in arrangement to differentiate the authority of a proven websites. Caller ID toolbar depends upon the passive visual indicator which changes in green which will represent a known good site, and when this indicator turns in to yellow it will shows a site with low risk, to colour red which will shows that the site is at high risk and must be the phished website. Here are examination of sites which are originated and created by country ,area of enrolment and blacklisted information. Caller ID Anti -phishing toolbar works under Microsoft Windows 98/NT/2000/XP with Internet Explore [2].

Cloud mark Anti-Phishing Toolbar:

This tool bar depends upon users rating. When user visits the site, the user has choice of describe the site whether it is genuine or not. With the help of cloud anti fraud system it will show the coloured image for each visited sites, green image will alerts the user that the site is visited or site has been real or not, red image indicates or will display the site has been fraud by the attackers or criminal, and the yellow icon will show the more details which is called to make a difference Fraudulent web sites are being blocked by the tool bar and the holder are authorized to the detailed page and the choice is given to stop or block the fake website or link.

The Cloud mark Anti-Fraud Toolbar runs under the Microsoft Windosws98/NT/2000/XP with Internet Explorer.

Earth Link Toolbar:

It is the combination of the user rating, heuristics and analysed verification. The toolbar gives permission t the holder to complain about the spoofed phishing sites in the tool. The sites have been checked and then inserted to the blacklist in case if it is fake and spoofed link. This tool also uses to examine the domain information like age, owner and the country. In this toolbar green thumb legitimate or represent the verified sites whereas grey thumbs up show the site is not reserved, but it is not confirmed, and red colour thumbs up shows the web site is fake. Where the yellow colour thumbs down mean the website is doubtful.

Earth link free software runs under Internet Explorer as well as Firefox [10].

There are also so many tools which are used as anti phishing tools like e-bay Toolbar the icon will display with alerted lights which will give indication about the site that how much safe the site is on the other hand other toolbars like Firefox2, Geo Trust Watch Toolbar, Microsoft Phishing Filter in Windows Internet Explore7, Net craft Anti-Phishing Toolbar, Netscape Browser 8.1, Spoof Guard.

How does one start phishing?

First the attacker will create his or her own fake websites, for example fake Face book web sites phishing. php file which will further collect the form of data and index. Html page. Then the attacker will go directly to the Face book page without logging in. Then the attacker will find the link and attacker will look forward action. For example section= <https://www.facebook.com/login.php> (This is the attempt one of the attackers).

The next step of attacker is he/she will create an account on free hosting websites like <http://www.mymy.com>. Next step is to upload the php file and html pages with his or her name, then the phishing website is created and the attacker can start phishing.

The main causes of phishing attack are as follows:

Today the most common and dangerous type of attack is phishing scams and the attack is faced by organizations. According to Verizon's Data Breach Digest found that 90% of all the data was breached by phishing. There are major 6 reasons of causing phishing which are as follows:-

There are major 7 reasons of causing phishing which are as follows:-

1. **User are the weakest link**– Users is weakest link of phishing because they would not be able to recognize or to spot the scams when the user receives one and by that point it's too late, with the victim already clicking links, like opening attachments and handling towards the user's name and the passwords. But the main positive point is weakness that organizations or individuals have the power to address. Currently IT departments are not at all confident in their own user to identify or to recognize in coming threats and related attacks.
2. **Insufficient back up process**- In ransom ware attack, most of the big organizations have insufficient back up process. Because of which it leaves them unable to quickly restore content on servers.
3. **Lack of user testing**- Most of the organization have no adequate procedures in order to test their users and leaving them unable to recognize about the attack.
4. **BYOD security risks**–Many organizations lack (Bring Your Own Device) policy where it makes easy to stay connected in with your employee. Basically, attackers will take the advantage of unsecured devices, networks and the malicious applications where the biggest security risks are hacking, malware and data leakage. There are major BYOD risks which are as follows.
5. **Shadow IT**– In this information technology is managed outside of the department of the IT companies. Where some users use low security products or infects the removable storage media which leads to data breaching.
6. **Lack of employee training**- Many employees in the company they may not fully understand the requirements of the company which will lead to compromising the security risks of the company.

7. **Poor mobile management**- When employee leaves the company for any reason how can the person be ensuring about the no longer mobile access of the employee therefore it become easy for the attacker to get back in to an app or system by using mobile access again.

Criminal organizations are well funded. Cyber criminals are shifting their focus towards the new tricks and the ways that how they can get or breach the details of the user or victim. Organizations are not doing enough the awareness of the staff it's not only the single step to protect the organization.

Major prevention from the phishing:

Spoofed messages and web contain few subtle mistakes, which includes spelling mistakes and the domain name Two factor authentication (2FA). It is the most effective method to find out the fraud attack where it contains (or add) the extra verification layer when user is logging into sensitive applications.

We can also use 2FA by two ways:

Something that user know (e.g., password and username)*

By using 2FA addition it should be strict password and with management policies. e.g., Need of new changed password and not to be allowed to reuse new password multiple times.

By giving some educational campaigns we can also get to know about the threat of phishing attacks by doing or enforcing secure practices (e.g., by not clicking on external links like emails and URL.)

Do not trust on such emails which will ask you to give the personal bank account information and OTP's.

Updating system with new security software like, anti-virus, firewall, spam filters and anti spyware.

Ignore the pop-up messages which will act as phishing rod for the phishers.

Current Statistics on Phishing Attacks:

- From 2020, according to the report of FBI phishing was most common type of cyber crime and it gets doubled in frequency nearly from 114,702 casein 2019 to 241,324 in 2020.
- Around 75% of the organization in the world experienced the phishing attack in 2020.
- Another 35% spear phishing attack.
- 65% of companies faced BEE attack.
- Where 96% of the phishing attacks was delivered by emails.
- 3% was happened by the malicious website and 1% via phone calls (which is known as vishing).

- 60% of the organization has lost their data.
- 52% of the organization had faced the account compromises.
- 29% of the organizations are affected by the malware attacks.
- 18% of the financial loses.
- 47% of ransom ware.

2. REVIEW OF LITERATURE

According to Phirashisha Syiemlieh, Golden Marry Khongnist, Usha Mary Sharma and Bobby Sharma witnessed the huge loss of money globally but the most Dreadful loss by the common users their personal information are being used against them for fraudulent acts even their bank accounts are being robbed without their concern by the fake information like winning lottery, reservations of hotel at cheap rates. According to author phishing is basically described as semantic attack where victims are being triggered by the attackers to give their bank information's.

There are many types of anti – phishing tool.

1. Cloud mark anti fraud toolbar
2. Earth link toolbar
3. e Bay toolbar
4. Geo trust watch toolbars

According to the Junior Professor “Philip Craiger”, Paul K. Burke “&” Chris S. Marberry “they said that investigation of the fake or fraudulent websites and links or malware can be detected by open – source tools. First examiner should be would be capable to recognize the tool which further match with the requirements and authenticate these tools under Daubert Standards. Second many tools runs under UNIX-like system (Free BSD, Linux, MacOS X,etc.)

According to Aakansha Tewari & A.K. Jain & B.B. Gupta they discussed about recent developments and protection mechanism against variety of attacks and the scope of future of results.

According to the Author “ Jyoti Chikkra” she discussed about the false emails which are looking legitimate and even the web pages and websites where users are asked to enter about their personal information which may look clear and she also gave the example that phishing is similar looks like fishing in a lake, she also explained that how internet has changed life human life where internet increased the human comfort in life and on the other hand it also increases the need for security measures too. Where basically the phishing is an online form of identity theft.

She also explained about the phishing theft how the things will appear–attacker will create fake emails and it has its own email address and the email will be delivered to the ISP and then they will get transferred to the mail program–from where the attacker they will get enter in to the user’s site in case when the user will click on or responses over the link. She took the Net Craft web server survey of 2013 and noticed about the all over phishing attacks which raised about 73% overall.

“Akarshita Shankar”, “Ramesh Shetty” and “Badari Nath K”– According to them, they had talked about the phishing attacks which are going very serious day by day according to them there are several methods that can used to phish and also used for the detection and avoidance of phishing attacks on the system. This study provides the in-sight to phishing and the basic mechanism about the attack.

According to them they had done evaluation on the constant growth of email user, client server and anti-spam filters are being used to find the variations in the appearance of the phishing scams. They also explained about the URL control features which will be capable to identify the connection of an address in the mails. They also aimed that in future the surface of text mining by enhancing the spam keywords in information’s. And it is also aimed to obtain more fine and correct results and the basic clarification by the help of duplicate l neural networks.

“Rami M. Mohammad”, Fadi Thabtah, “ Lee Mc Clusky they explained about the phishing counter measures by legal solutions, education, technical solutions and instantaneous based protection approach. According to them the site of web sites is constantly changing, self-structuring and interactive ways in response to the change in environment which characterize the websites.

Author “Dr. Radha Damodaram” she explained about the browser integrated tools, eset security, using anti phish and Dom techniques where by the set of techniques she talked about the detail examinations of various ways of operating and handling of the system which includes start up, registry content and network connections. She used the technology to discover hidden objects (root kits). SysInspector signs are verified in XML files and it can be presented to IT experts for the future examination.

3. MATERIALS AND METHODS

Methodology

By using and some techniques, tools and awareness we can detect and analyze the tool. There are different types of ways which use to detect the web pages as a phished website.

In article “An Examination Of Anti- Phishing Toolbars” the authors used so many tools to detect the phishing according to the previous study it determined that the area to which why the holder fall for the phishing scams in which user can ignore the information which is provided by the anti phishing toolbar. In this article they told and wrote about that how the tools have been used to detect and ignores the scammed websites and pages. In this article paper the authors has managed the three contributions, first is in which they designed and implemented the examination bed for digitally determination of the anti phishing toolbar, second the research of result experiments which assess the ability of ten anti phishing to operating the system which uses different methods to cipher and to detect the phishing websites. Third where the authors described the techniques for circumventing each toolbar tested.

There are different types of methods which can be used in the deception of the phishing websites and the scams. The toolbars in this research article examined the study of the different combinations of the methods and observing by using every type of operating system r to get general alignments about the process of the toolbars.

Cloud mark Anti- Fraud Toolbar: This tool can be used as the Anti- spam, Anti- Phishing and Anti-Virus, where the headquarters are situated in ‘San

Francisco, CA and USA. Cloud mark toolbar takes zero hour response, real time protection for all the messaging threats. This tool blocks the attack within or in between 20 seconds to 3 minutes. Advanced fingerprinting algorithms are also used in precise identification of electronic mail with a single fingerprint whereas the Trust Evaluation System (TeS) use to track the reputation of each integrity of data and accuracy it also determines when to mark the of identify the fingerprint spam, phishing, spam and virus.

When the user visits the website, holder has the choice to report the website whether it is genuine or not. The operating system will display the colours images for each website which person had been entered previously.

Green coloured image indicates the website is standardized as authorized site, where red icons which will indicate the site has been determines as fake and at last the yellow image will tell that nor required details is called up on the web site to make the examination. The ratings which was given in the site by the user have been computed and for further the user can use the site by looking over the ratings.

The cloud mark anti- fraud operating system works on the ‘Microsoft Windows 98/NT/2000/NP with the Internet Explorer.



Figure 1: The Cloudmark Anti-Fraud Toolbar indicating a legitimate site.

Earth link toolbar: With the togetherness of heuristics, holders views and the written authentication, these small details about the URL and link will tell that this site is genuine or not where the earth link toolbar gives permission to the user to report the suspected phishing website or link and then these websites can be further being checked and inserted to the black list, it can examine the domain name registration information.

The toolbar will show the thumbs up which swap its colour and place.

Green coloured thumb shows that the site is legitimate and verified where the grey colour thumb shows that the website is not original or real, but it is not concluded.

The red colour image thumbs down will alert the user and tells that the site is fraudulent and the yellow thumb down will mean that the website is doubtful.

Earthlink operating system works under the ‘Internet Explorer as well as Firefox {10}.



Figure 2: The EarthLink Toolbar indicating a legitimate site.

eBay Toolbar: In this tool account guard indicators has three different types of modes: green, red and grey. The icon will display green in the background when the user visits in the genuine website and the red colour will display in the back side when the website is called to be phished websites. The image will display in grey when the website is not handled by the eBay system the website is called to be phished.

This toolbar also gives the user to report the site if the user finds that the site is not genuine eBay operating system or application works on the 'Microsoft Windows 98/ME/NT/2000/XP with the Internet Explorer'.

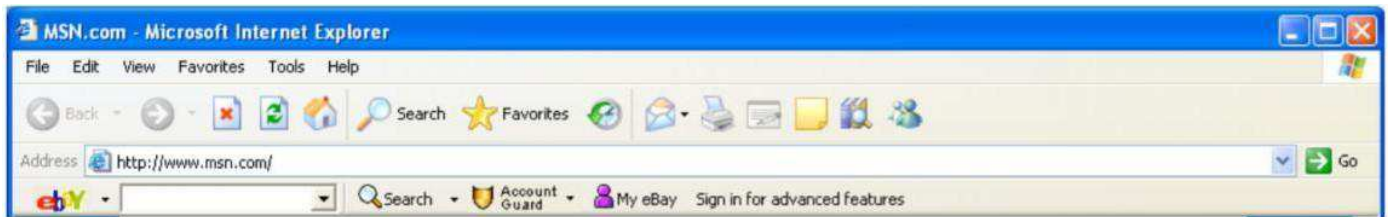


Figure 3: The eBay Toolbar at a site not owned by eBay that is not known to be a phishing site.

Geo Trust Watch System: This tool works under several reputation services and certification authorities to detect and to analyze the entrusted sites. This tool also works under three signals light (green which indicates the trusted and verified site, yellow which is not verified or red which is verified as fraudulent).

The toolbar gives permission to the user to store custom images which is constantly displayed so that user can know that this operating system is not being copied or spoofed. This toolbar works under the "Microsoft Windows 98/NT/2000/XP with Internet Explorer". Trust Watch determines the about the site that is this fraudulent or not in which the suspected company compiles a blacklist which included the site reported by the user through the button provided by the toolbar.

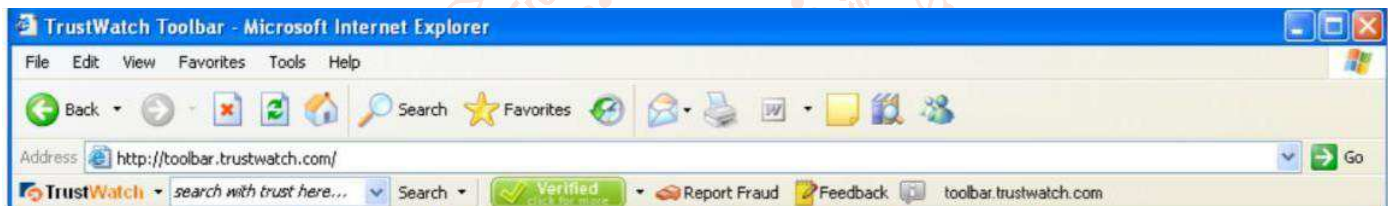


Figure 4: The GeoTrust TrustWatch Toolbar at a verified site.

Google Safe Browsing: It is designed to identify the fraudulent websites and the privacy of Google browsing basically was really a Firefox Extension and currently this is integrated into full toolbar which is built into Firefox 2.0. and gives the source cipher for the secure browsing feature and check URL's against a blacklist.

When site get downloaded the tool get combines with advanced technologies and algorithm with project about misleading page from the number of sources whereas the conclusion is this tool uses blacklist as well as heuristics.

The toolbar also includes the page rank. It popup out when the site is suspected fraudulent which alerts the user and the user will leave and ignore the site. The toolbar also provides the up to date protection by sending URL's sites. This can runs under the 'Microsoft Internet Explorer under the Windows XP/2000 SP3+, or Firefox on most of the platforms'.

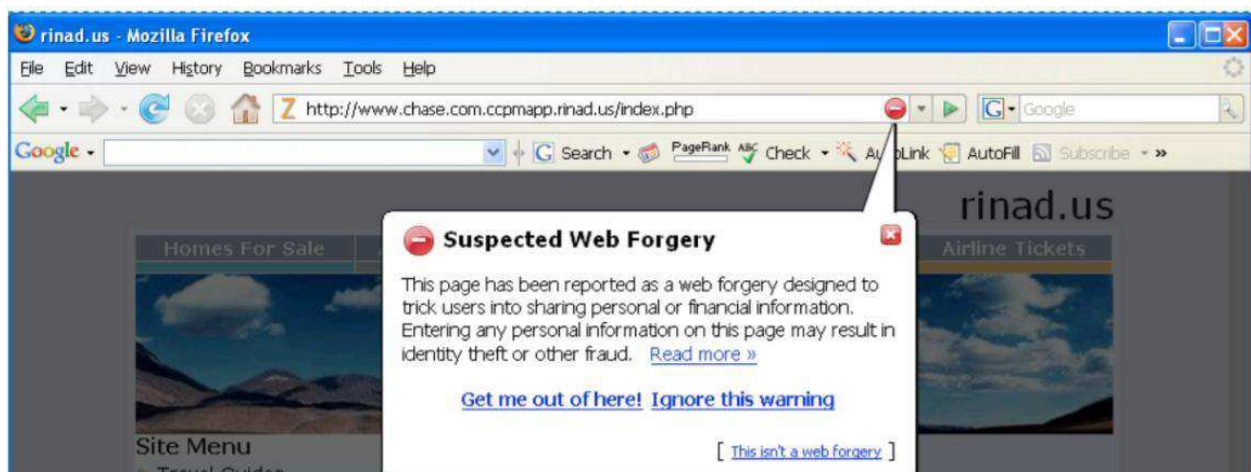


Figure 5: The Google Toolbar at a fraudulent site.

McAfee Site Advisor: “Works and runs under Microsoft Windows, Linux and Mac OS X with the Firefox Web Browser and Internet Explorer under Windows “.

The toolbar follows the traffic light model where green indicator indicates that the website has determined to be good; but red colour indicator points some serious issue and other yellow colour points that the site may have some small issues had in the past and it remains grey when the site information is not present.

The above given determination is done by the manual verification and automated heuristics. Where the toolbar will examine the domain (name age, date of birth etc.) registration and the user reviews.

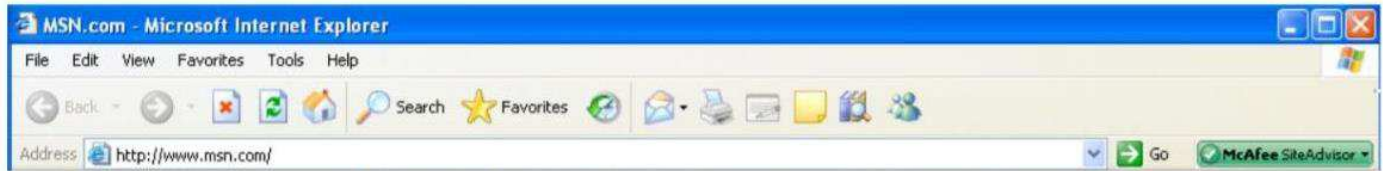


Figure 6: McAfee SiteAdvisor at a verified good site.

Microsoft Phishing Filter in Window Internet Explorer 7: This toolbar runs under the “Microsoft Internet Explorer 7 web browser” which built in the phished filters and basically this toolbar is depend on the blacklist which is hosted by the Microsoft.

When the user comes in contact with the suspicious or suspected phishing website the user is directly redirected to the warning message and asked that they want to visit the site or close the window.

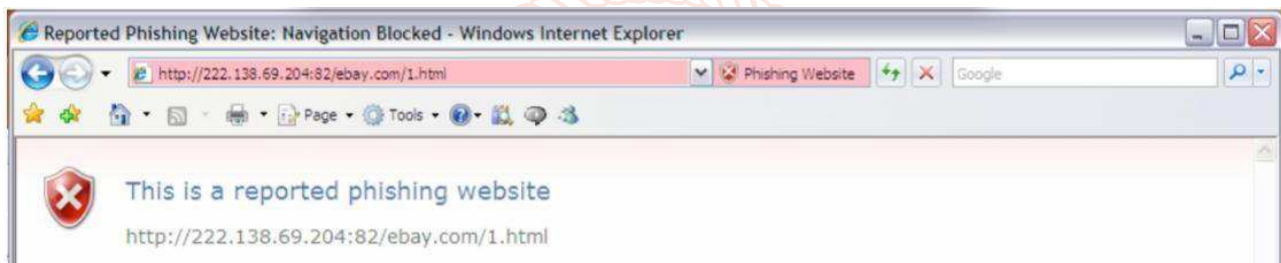


Figure 7: The Microsoft Phishing Filter in Windows Internet Explorer 7 at a fraudulent web site.

Net craft Anti Phishing Toolbar: This toolbar basically runs under ‘Firefox 1.0 and Microsoft Internet Explorer under Window 2000/XP.

It traps suspicious URL’s, and apply to show of the browser map reading controls in all the windows, in order to protect against the popup window that attempts to hide the reading controls measures, and it directly shows to the websites by clarifying the location.

The net craft toolbar will block list the site o the basis of the reviews and feedback exempted by users.

When holder tries to generate the website which present on the black list then that popup warning will warn the user which cancels the access of the user on that particular black listed websites, operating system also displays the viewer between one to ten ratings and also as at the hosting navigation of the website.

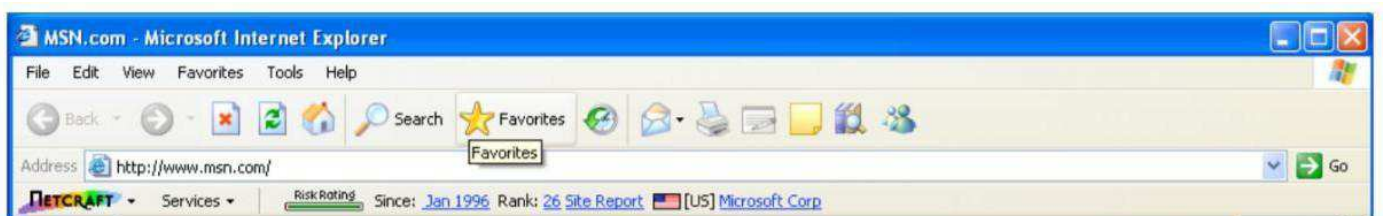


Figure 8: The Netcraft Anti-Phishing Toolbar at a legitimate web site.

Netscape Browser 8.1: This toolbar runs under the Microsoft Windows, Linux and Mac OS X. It also includes phishing filters, or it also relies on the black list which can be set up by the AOL and reconditioned frequently, and when the suspects gets enters inside the phishing site which redirected to built the user in warning page, where the user are shown to the real or original URL.

Spoof Guard: This runs under‘Microsoft Windows 98/NT/2000/XP but with the Internet Explorer’.

Spoof guard does not use white and black lists instead of this the tool employs a sequence of profiling to know the phished sites.The very first step is to check the name of the domain and the comparison is done with the

websites which has been freshly entered by the user or holder, in order to catch the fraudulent websites that have domain name similar.

Next the URL is examined and detects the confusion as well as non pointed port numbers.

After this the appeal on the webpage has been examined or goes through by looking over the record of making password fields, lodge links and the icons.

Spoof guard computes the outcome for each and every webpage in the type of laded score of the outcome of each set of profiling, if the outcome exceeds a particular entrance, the tool will show red image which means the website is fully phished, when the image turns in colour yellow which indicates that the site cannot be examined about the websites, when it turns into the green it indicates the site is safe to use.



Figure 10: SpoofGuard at a legitimate web site.

There are some experiments and analysis step which are performed by the authors and are as follows:

Get potential phishing sites from phishing feeds. Task manager will download the URLs from the phishing feeds. To obtain the catalogue of the phished websites the initial is to take out the URLs from the phishing electronic mails. Where phished electronic mails contains link to genuine and fake web sites, then it manually confirms that remaining URLs are scams.

Heuristics will remove first URLs which points to the image (as against to HTML pages) and the websites that do not contain any written file entry fields which appealed the tactful information like passwords, name, age, work and credit card numbers.

Second is profiling which uses as variant of Robust Hyperlinks which removes the sites which are legitimate. The word contains the linguistically signatures and then cater into the search engine.

The matches are important in the recent web page and the designation name which is on the ridge look result.

Testing methodology: Basically the examining anti- phishing toolbars will take too much time and very hard to proceed, and the final outcome which should be matched, with the multiple tools need to be examined on the same of the uniform resource locator within the short period of time.

The toolbar need to examined with URLs withdraw from the phished messages after that the arrival on the holder mail boxes. All the toolbars examined appear to have some problems. Anti – fraud toolbar could help to examine all the fake web sites without any incorrect positive result, but if it has usability problems user may still fall to fraud. In upcoming days, or in future the technically sound tools is of little use that what is trying to communicate them, it would be worth to know that the user examination still have many solutions which are already popular in use.

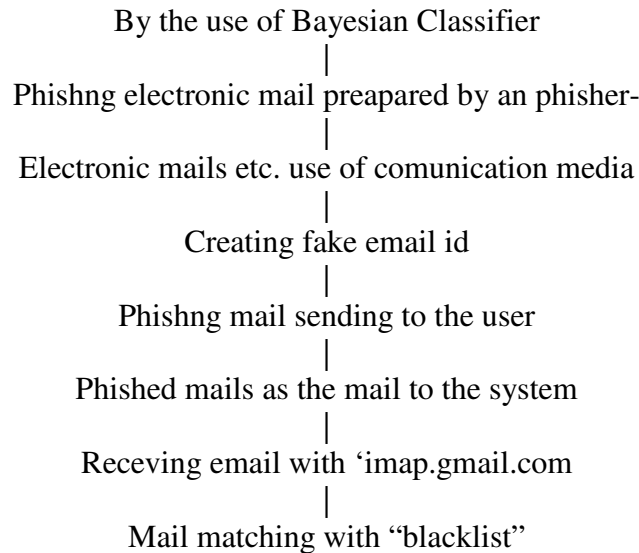
How the user reacts to the danger or online threat as well as, how the user will react with the false positive.

In another article the author told about the Anti- Phishing Simulator:

The Bayesian Classification algorithm is used in the database. It instantly works on spam messages to those words that have stimulating clause like shopping, lottery etc. The page HTML code with the URL control.

The main aim and work of this simulator is to command the safety of details of the user and to prevent violations to check that spam is present on the recent data or not.

The below given flow chart with the direction will tell that how the actual work happens.



The rapid extension of email holders has calculated in emails which are becoming so advanced. Anti- spamming filters are use to pin point the varieties features of spam emails, many effective methods and shortcuts are being enhanced with the insertion of spam senders data as digital images, ppts, pdf and words.

Anti- phishing system gathers the phished mails and spammed messages at mutual point. It allows the user to handle the spam box whenever the user wants it. In future it is aimed to detect with the ground content mining is growing rapidly by the spam keyword detailed information, it also focuses to get the most excepted outcome and differentiation with the artificial neural networks.

4. RESULT AND DISCUSSION

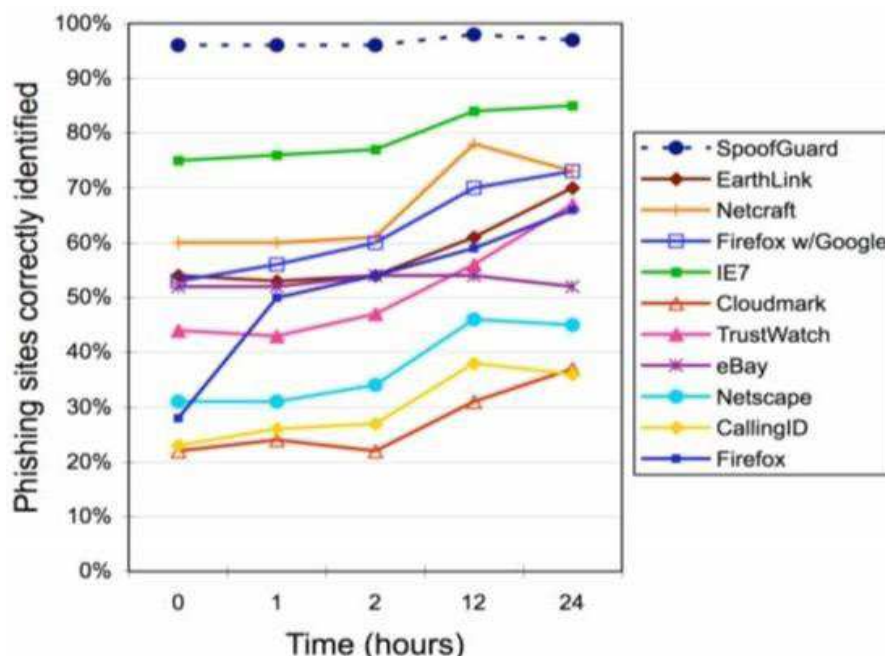
Result:

According to many research articles the anti-phishing toolbar which were examined in the study, for example- spoof guard anti- phishing toolbar which is best for the identification of the fraudulent and phished web sites and is also known to make the no use of the blacklists with the help of heuristics toolbar also allow to detect the spoofed websites, by it had very high false positive rate, on the other side the other toolbars like 'Earth link, Google, Net craft, Cloud mark, IE7, identified most of the fraudulent websites correctly but few are false positive like these tools missed 19% of fraudulent websites and the performance of other four toolbar are correct – which identifies less than half fraudulent sites.

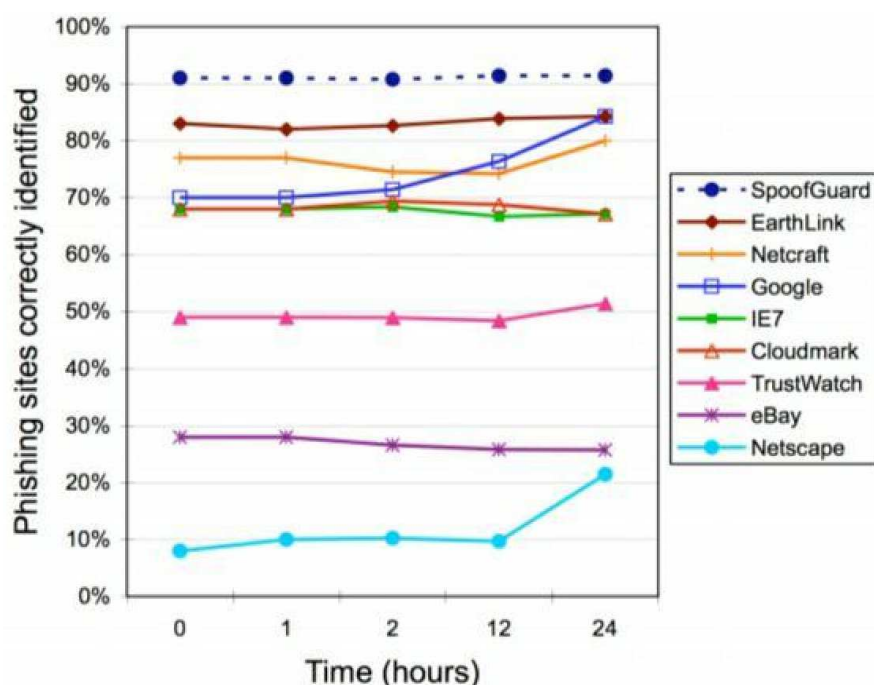
Most of the system was also capable to recognize the phishing websites. In this study with the help of the combination of the technologies and toolbars the phished websites can be determined. In order to get the comparable result we need multiple toolbars which should tested on the equivalent set of the uniform resource locators with less duration of time.

Toolbars or operating system should be examined with the uniform resource locators bring out the messages right away when it arrives in the mailbox of the user.

Researchers also observed some variations in result on the basis of the use of the toolbars APWG feed larger fraction of spoofed websites and by the use of feeds captures moved to their blacklists not in the tool recognized all the phishing URLs links.



Graph 1.



Graph 2.

Discussion:

Overall the anti-phishing operating system was analysed in the above given statements and according that which a lot of work to be desired. One of the best toolbar is Spoof Guard which did very great job in the identification of the fraudulent websites, but this toolbar also performed some false positive result and recognizes a huge number of fraction of genuine fake websites and on other side or hand, the another type of toolbars like Earth Link, Google, Net craft, cloud mark, and IE7 were able to find out the most fraudulent websites and have also false positive results because they missed more than fifteen percent of fake websites and the other tools will positively recognized the fake websites and sometimes few tools are not able to identify the fraudulent websites.

There is no single technique which will always give accurate result for the identification of the phishing websites for example many of the system which are used blacklists yet just half of the tools were capable to recognize the fake websites. Sometimes the tools can also use profiling which will give permission to recognize the phished websites which yet not been put into the black lists and on the other hand the only toolbar tested is known no use of blacklist which was spoof guard. Spoof guard may also be improved through the use of white lists, will prevent the problems which occur when the phishing websites when user visits before their alternative genuine websites.

By examination methods – anti phishing toolbar takes more time to analyze and not easy procedure and to get the complete compatible results many types of operating systems need to be checked on the equal set of the uniform resource locator at the short period of time and the uniform resource locators are the only useful for the testing and detecting purpose, sometimes testing URLs are very problematic. Most of the operating system use the results as input in their devices (blacklists) however not even a single toolbar is capable to identify the phishing websites properly. There were lots of differences which were observed because of the results and how much the alterations in the tools on the operating system and the procedure maintaining for their blacklists.

There are many other toolbars which is directly interacts with the user. There are eight of the ten toolbars examined and indicates the user on the bases if the colours where green colour shows the genuine website and red colour shows the correctly recognizes phished websites and other seven toolbar used yellow colour or gray colour image shows that no result is satisfied about the identified site.

Coloured indicators uses pops- up the dialog boxes warns when the website recognized the fake websites. While the other toolbars blocks the phishing sites unless the user overrides the block. According to the previous studies shows that presented with dialog boxes which contains button which automatically dismiss them, many suser use to dismiss the boxes without reading from where the user will get phished.

5. SUMMARY AND CONCLUSION

Summary:

The word phishing which is very unique and basically derived from the analogy of early internet criminals where it means - “fish for passwords and for financial data” from the unsuspecting internet users. Or basically the phishing refers to the process of breaching the personal information like bank details, card number and passwords of the user in order to gain some money and the attackers use to target the individual contacted them by the emails or telephones and by pretending legitimate institution which can result in identity theft and the financial loss. By using some techniques and by the combination of the tools we can detect and analyze the phished websites and can also learn about the prevention measures from the phishing attack. Spoofed messages and web contains few subtle mistakes, which include spelling mistakes or domain name, two factor authentications (2FA). It is the most effective method to find out the fraud or attack where it contains (or add) the extra verification layer when user is logging in to sensitive applications. Do not trust on such emails which will ask you to

give the personal bank account information and OTP's, updating system with new security software like, antivirus, firewall, spam filters and anti spyware, ignore the pop up messages which will act s phishing rod for the attackers.

Conclusion:

On the basis of experiment assessment the effects of five phishing toolbars, by the evaluation of results of large data sets across the long period of time, and they found that three of the ten toolbars like- Spoof Guard, Earth link and Net craft these three toolbar are correctly capable to recognize the 75 percent of the fake and phished sites positively during same time period, Spoof Guard mistakenly found the 38 percent of the authorized Uniform Resource Locators. as phished Uniform Resource Locators. And by looking over this percentage of the error might nullify the benefits of the spoof guard offers in detecting the sites. By looking over all errors the final outcome comes out that more or advanced work is needed to do in the area of the operating system and operation of the tools.

While using the anti- phishing operating system sometimes bad use which could mean s the difference between correct steering someone away from the phishing site and having them ignore the alerts only comes out when the victim of the theft or eyewitness. And according to the study some interesting facts comes about the attacker which he/she would go in order to fulfil the desirable needs. Organizations are taking an initiative move of spreading awareness statement about the fake information's which will alert the user fro, getting robbed or phished.

The main goal of this study is to protect the big organizations and every common innocent people from the phishing attack by alerting them about the real facts and real way that how the attackers trap the users. By carefully going through the websites “sender is not valid valdosta.edu address “Admin Team” which do not match with email address @pugmarks.com which is not valid, subject line is in all capital letters and multiple exclamation which are trying to get the attention, highlighted “Click Here “by clicking on this external sites will start steal your credentials and will install malicious software.

There is many other major problems which phases each person and every organization for example - Bank account breaching, Lack of awareness about phishing to the organizations ,Weakest links (users), Backup process is insufficient, Security risks, Organization of attackers are well funded, Phishing tools are at low cost, Malware is becoming sophisticated, Use of password multiple times.

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